

Figure 12-3. *Resistors*

In addition to resistors of the various types and sizes shown ranging from one-sixteenth of a watt to a massive 25-watt aluminum resistor, this image also shows several types of variable resistors. These resistors change their resistance depending on what kind of resistor they are. For example, a photoresistor or photocell will change its resistance depending on how light or dark it is, or how much or how little light hits the resistor. A thermistor changes resistance based on its temperature. A force sensitive resistor will respond to the amount of pressure or force applied to it. A special kind of resistor called a potentiometer will change resistance based on the position of a knob or slider.

We will use resistors for all manner of things, from limiting the current to other devices, reducing the voltage in a circuit, to measuring the change of variable resistors.

Capacitors

A capacitor will store an electrical charge when current is flowing into it only to release this charge when the current has been removed. Capacitors come in many different shapes and sizes, as shown in Figure 12-4, and are made from many different materials, but they all pretty much do the same thing. Their amount of capacitance is rated in a ridiculously large unit called a farad, although our capacitors will be a lot smaller than this, ranging from microfarads (µF) or a millionth of a farad, nanofarads (nF) a billionth of a farad, and picofarads (pF) a trillionth of a farad. These are small values indeed.